US 20250273305A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0273305 A1
(43) Pub. Date: Aug. 28, 2025
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(54) COMPUTER-READABLE RECORDING MEDIUM STORING ALGORITHM SELECTION PROGRAM, ALGORITHM SELECTION METHOD, AND INFORMATION PROCESSING APPARATUS

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(21) Appl. No.: **18/985,715**

(22) Filed: **Dec. 18, 2024**

(30) **Foreign Application Priority Data**

Feb. 28, 2024 (JP) 2024-029192

Publication Classification

(51) **Int. Cl.**
G16C 20/20 (2019.01)

(52) **U.S. Cl.**
CPC *G16C 20/20* (2019.02)

(57) **ABSTRACT**

A non-transitory computer-readable recording medium stores an algorithm selection program for causing a computer to execute a process including: in a case where a solution for structure optimization of a molecule is calculated using a quantum chemical calculation algorithm that repeatedly executes calculation for a plurality of cycles, executing a first quantum chemical calculation algorithm for a target molecule and calculating a nuclear gradient norm for each cycle; executing the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is equal to or more than a first threshold; and executing a second quantum chemical calculation algorithm of which execution time is longer than the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the first threshold.

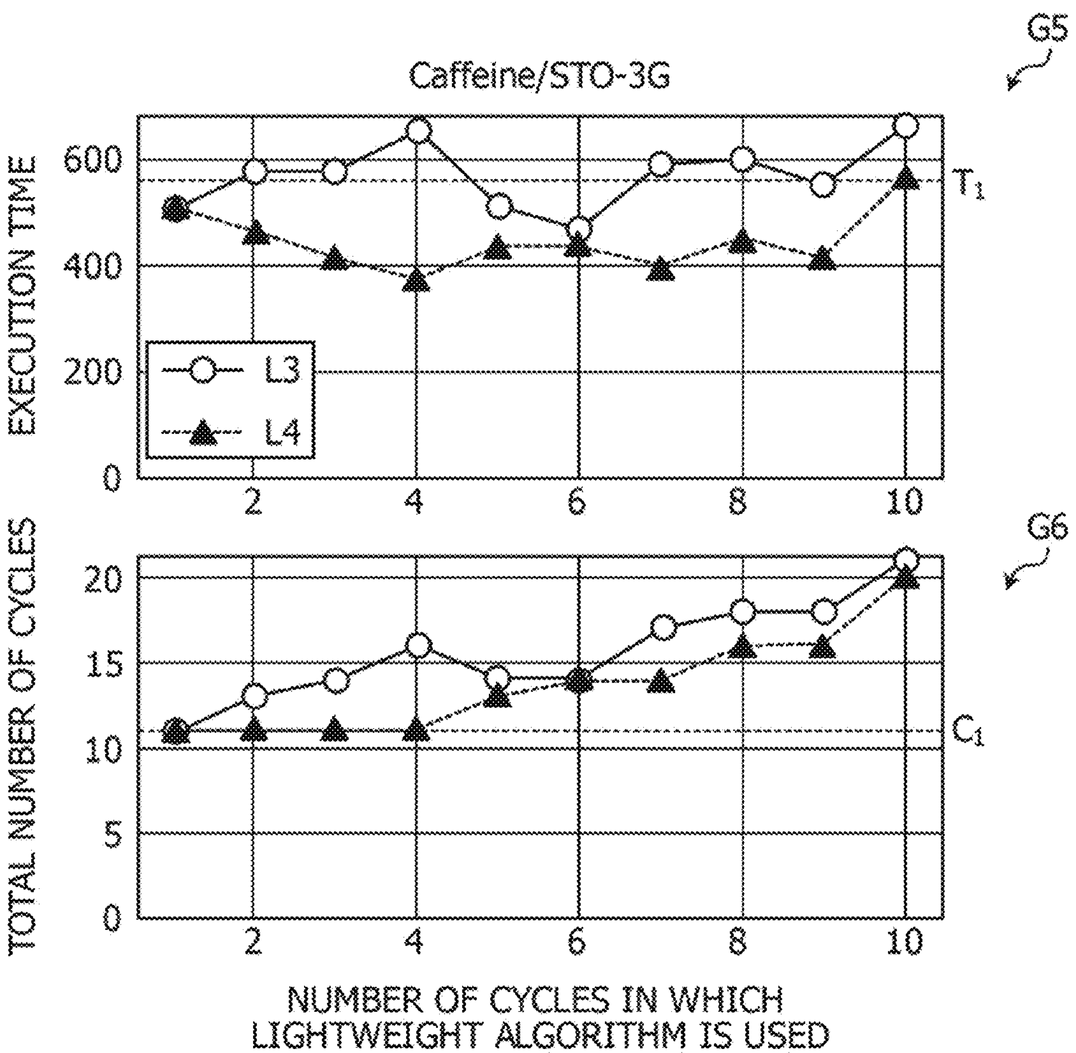


FIG. 1

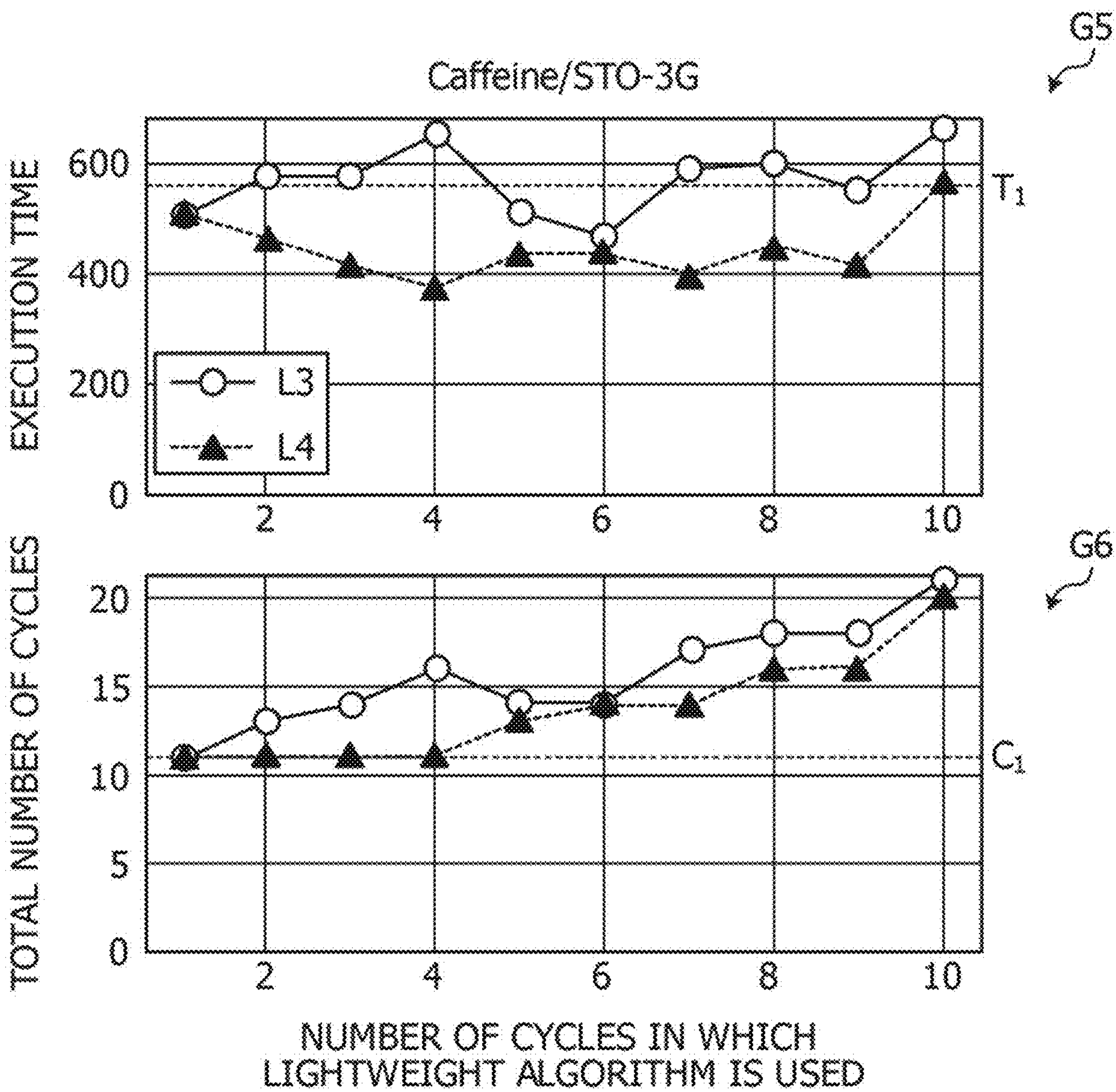

G5
Caffeine/STO-3G
EXECUTION TIME
600
400
200
0
L3
L4
T_1
2
4
6
8
10
G6
TOTAL NUMBER OF CYCLES
20
15
10
5
0
C_1
2
4
6
8
10
NUMBER OF CYCLES IN WHICH
LIGHTWEIGHT ALGORITHM IS USED

FIG. 2

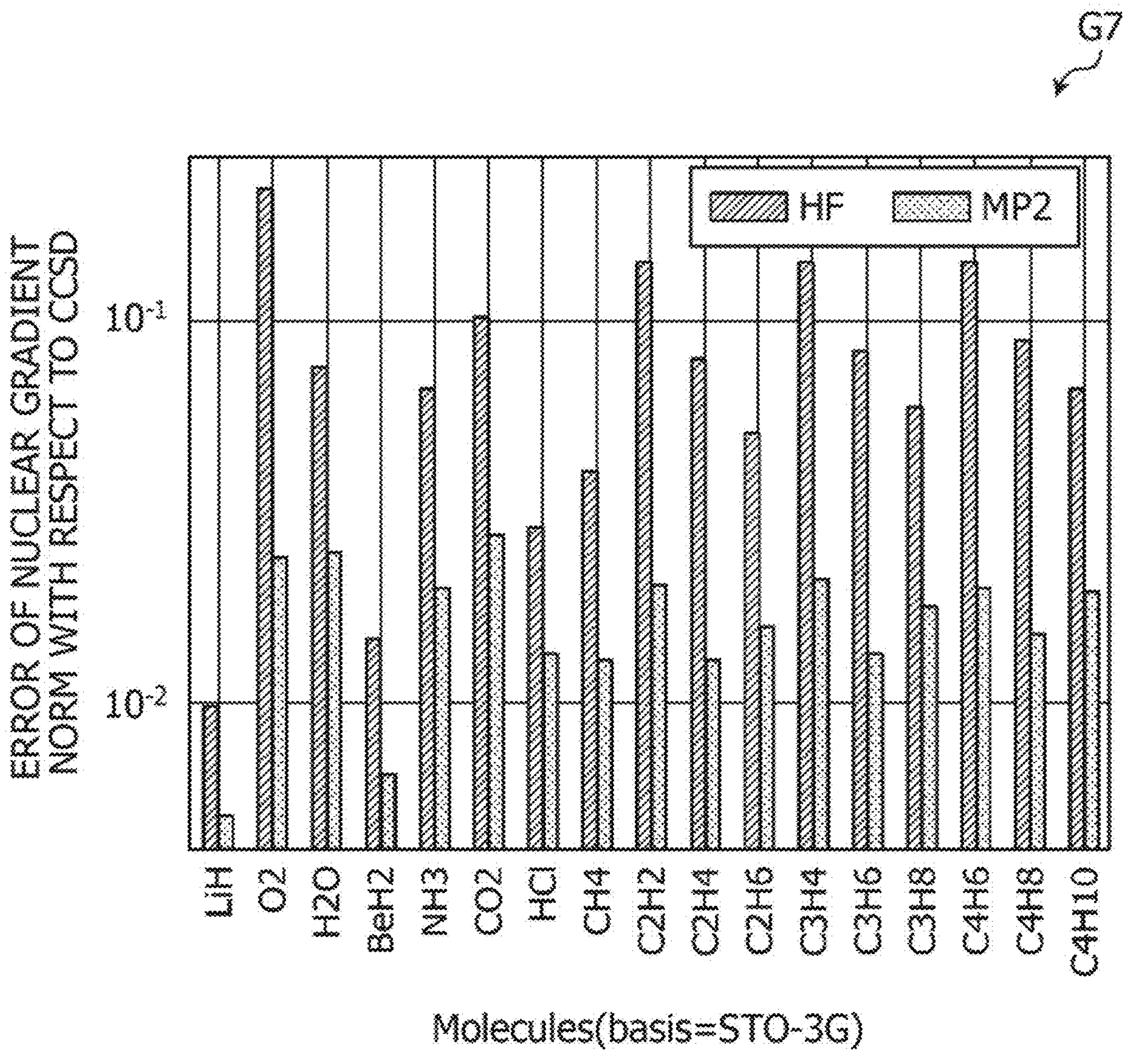

G7
HF
MP2
ERROR OF NUCLEAR GRADIENT NORM WITH RESPECT TO CCSD
10^-1
10^-2
LiH
O2
H2O
BeH2
NH3
CO2
HCl
CH4
C2H2
C2H4
C2H6
C3H4
C3H6
C3H8
C4H6
C4H8
C4H10
Molecules(basis=STO-3G)

FIG. 3A

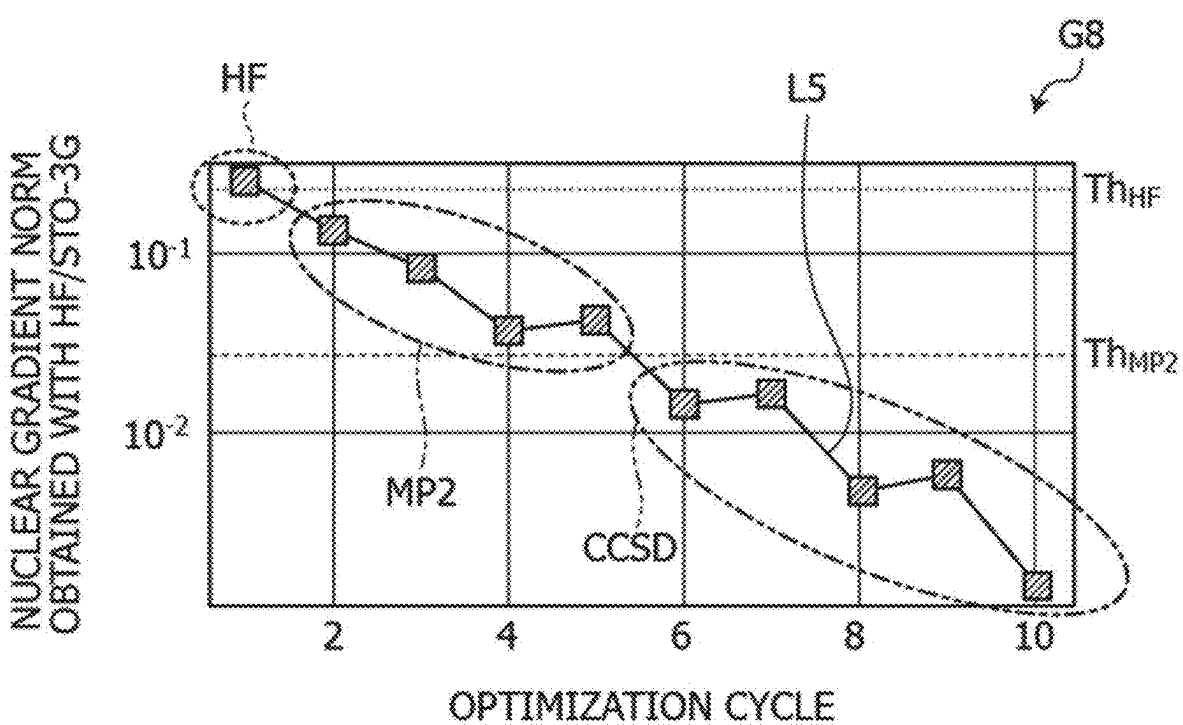

G8
HF
LS
Th_{HF}
Th_{MP2}
10^{-1}
10^{-2}
MP2
CCSD
NUCLEAR GRADIENT NORM OBTAINED WITH HF/STO-3G
2
4
6
8
10
OPTIMIZATION CYCLE

FIG. 3B

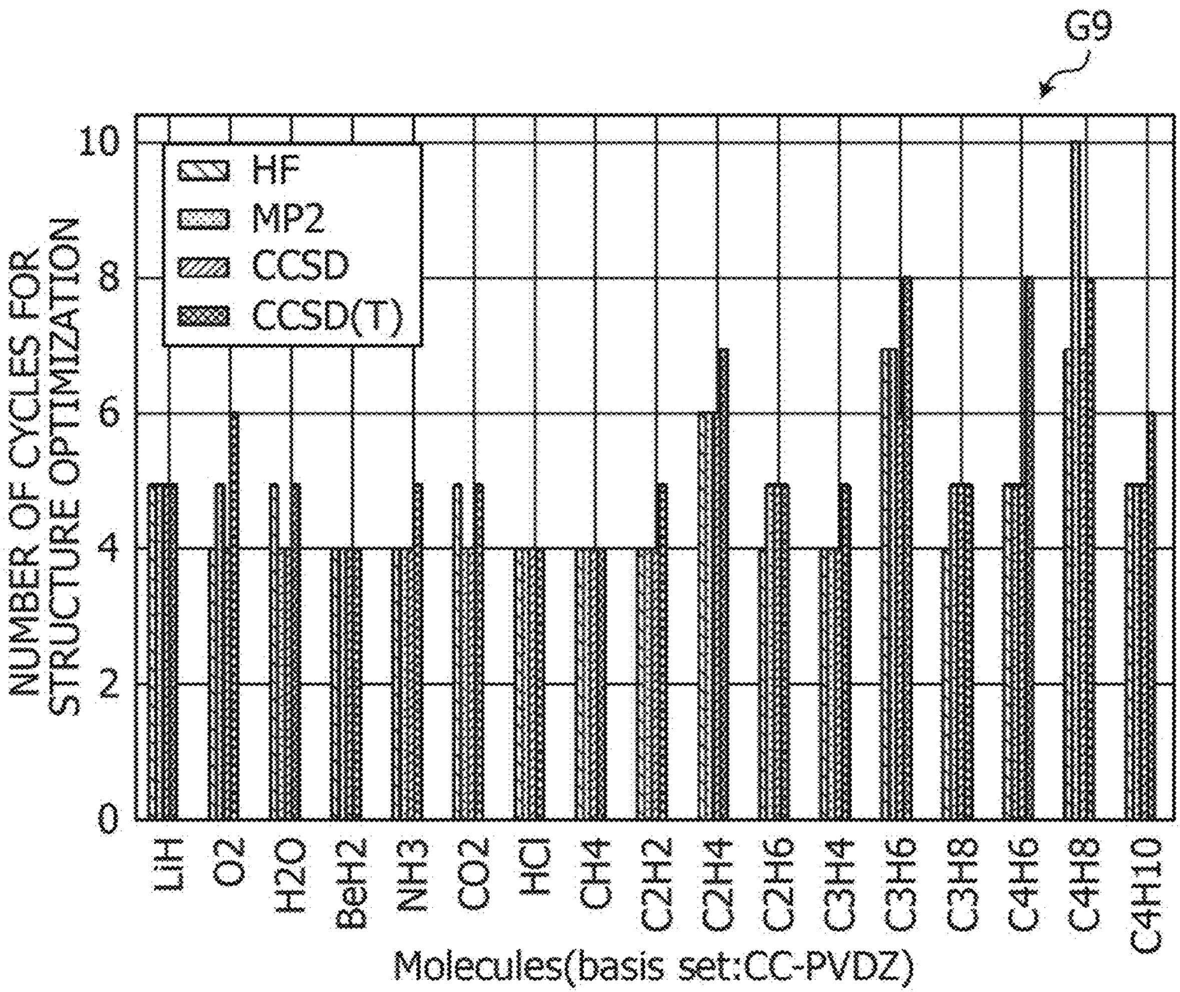

G9
HF
MP2
CCSD
CCSD(T)
NUMBER OF CYCLES FOR STRUCTURE OPTIMIZATION
0
2
4
6
8
10
LiH
O2
H2O
BeH2
NH3
CO2
HCl
CH4
C2H2
C2H4
C2H6
C3H4
C3H6
C3H8
C4H6
C4H8
C4H10
Molecules(basis set:CC-PVDZ)

FIG. 4

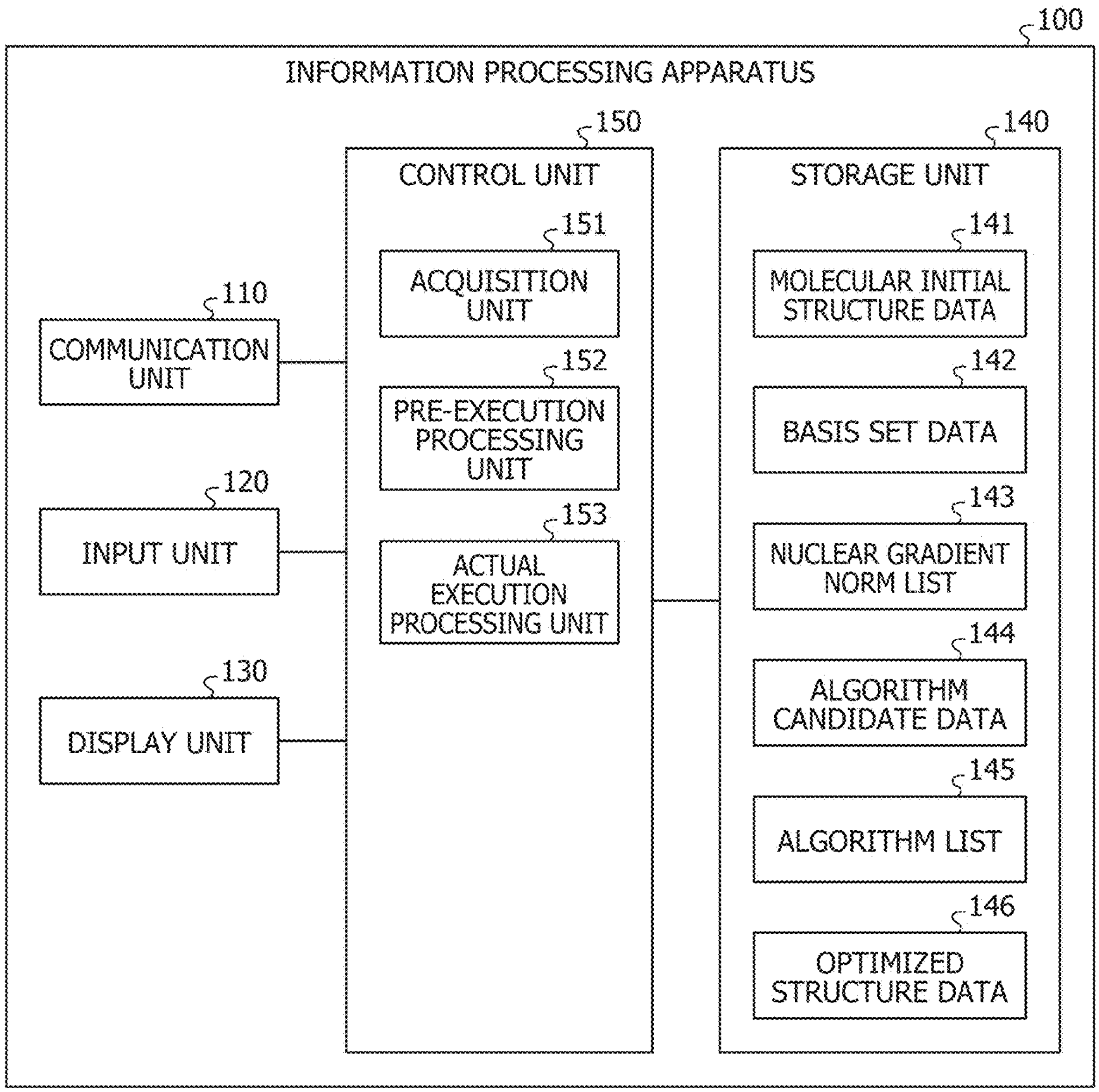

100
INFORMATION PROCESSING APPARATUS
150
CONTROL UNIT
151
ACQUISITION UNIT
152
PRE-EXECUTION PROCESSING UNIT
153
ACTUAL EXECUTION PROCESSING UNIT
140
STORAGE UNIT
141
MOLECULAR INITIAL STRUCTURE DATA
142
BASIS SET DATA
143
NUCLEAR GRADIENT NORM LIST
144
ALGORITHM CANDIDATE DATA
145
ALGORITHM LIST
146
OPTIMIZED STRUCTURE DATA
110
COMMUNICATION UNIT
120
INPUT UNIT
130
DISPLAY UNIT

FIG. 5

144

ALGORITHM	THRESHOLD
HF	Th_{HF}
MP2	Th_{MP2}
CCSD	Th_{CCSD}

FIG. 6

145

CYCLE NUMBER	ALGORITHM
1	HF
2	MP2
3	MP2
4	MP2
5	MP2
6	CCSD
7	CCSD
8	CCSD
9	CCSD
10	CCSD
11	CCSD

FIG. 7

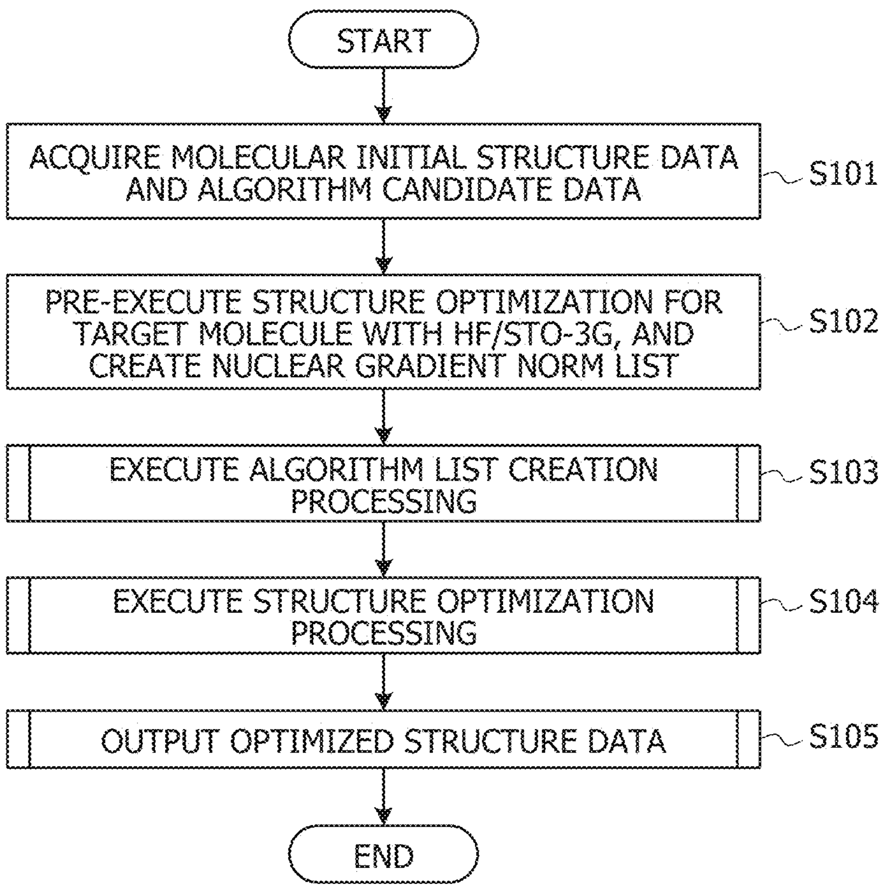

START
ACQUIRE MOLECULAR INITIAL STRUCTURE DATA AND ALGORITHM CANDIDATE DATA
S101
PRE-EXECUTE STRUCTURE OPTIMIZATION FOR TARGET MOLECULE WITH HF/STO-3G, AND CREATE NUCLEAR GRADIENT NORM LIST
S102
EXECUTE ALGORITHM LIST CREATION PROCESSING
S103
EXECUTE STRUCTURE OPTIMIZATION PROCESSING
S104
OUTPUT OPTIMIZED STRUCTURE DATA
S105
END

FIG. 8

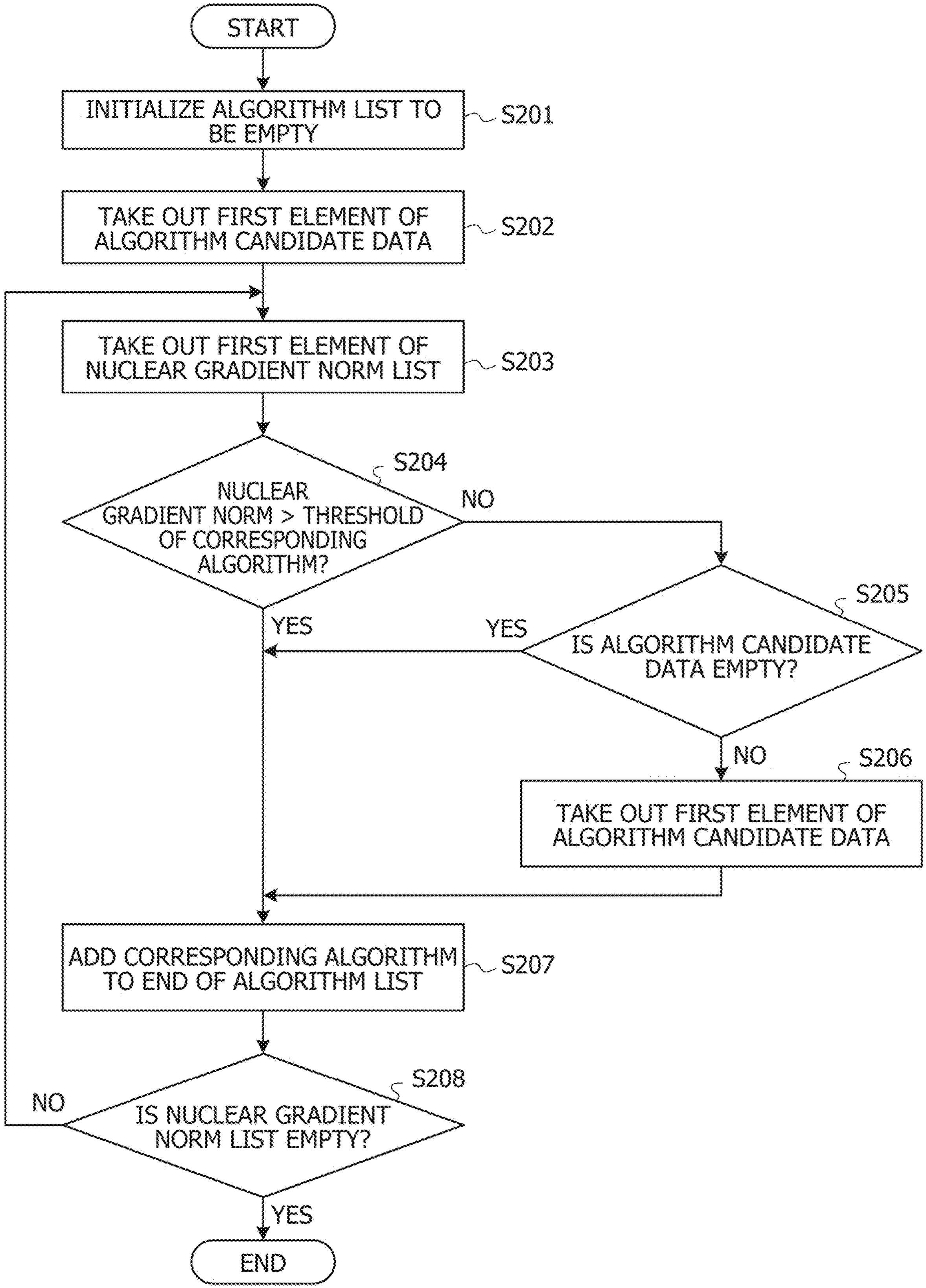

START
INITIALIZE ALGORITHM LIST TO BE EMPTY
S201
TAKE OUT FIRST ELEMENT OF ALGORITHM CANDIDATE DATA
S202
TAKE OUT FIRST ELEMENT OF NUCLEAR GRADIENT NORM LIST
S203
NUCLEAR GRADIENT NORM > THRESHOLD OF CORRESPONDING ALGORITHM?
S204
NO
YES
IS ALGORITHM CANDIDATE DATA EMPTY?
S205
YES
NO
TAKE OUT FIRST ELEMENT OF ALGORITHM CANDIDATE DATA
S206
ADD CORRESPONDING ALGORITHM TO END OF ALGORITHM LIST
S207
IS NUCLEAR GRADIENT NORM LIST EMPTY?
S208
NO
YES
END

FIG. 9

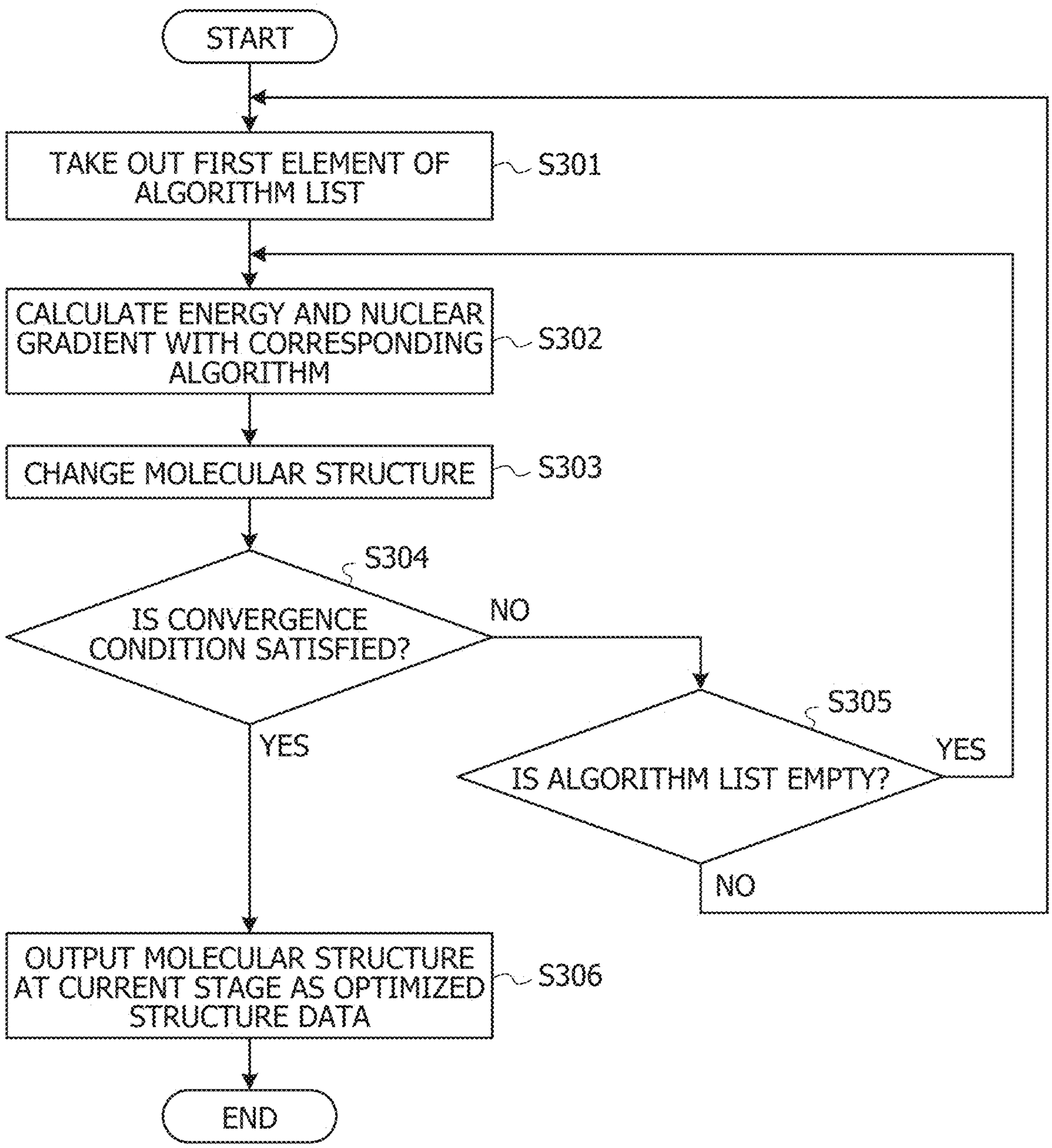
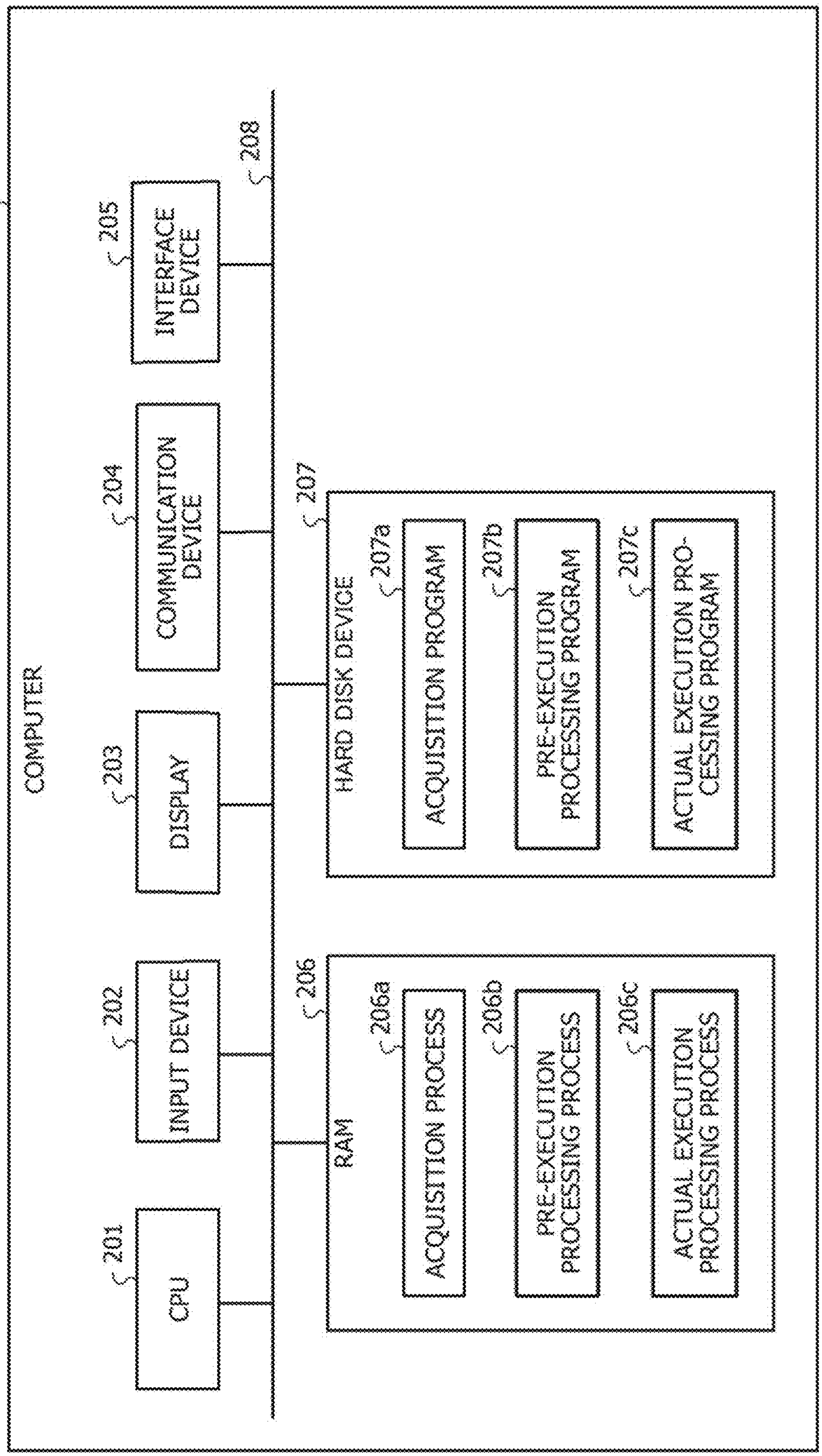

START
TAKE OUT FIRST ELEMENT OF ALGORITHM LIST
S301
CALCULATE ENERGY AND NUCLEAR GRADIENT WITH CORRESPONDING ALGORITHM
S302
CHANGE MOLECULAR STRUCTURE
S303
S304
IS CONVERGENCE CONDITION SATISFIED?
NO
YES
S305
IS ALGORITHM LIST EMPTY?
YES
NO
OUTPUT MOLECULAR STRUCTURE AT CURRENT STAGE AS OPTIMIZED STRUCTURE DATA
S306
END

FIG. 10



COMPUTER
200
CPU
201
INPUT DEVICE
202
DISPLAY
203
COMMUNICATION DEVICE
204
INTERFACE DEVICE
205
208
RAM
206
ACQUISITION PROCESS
206a
PRE-EXECUTION PROCESSING PROCESS
206b
ACTUAL EXECUTION PROCESSING PROCESS
206c
HARD DISK DEVICE
207
ACQUISITION PROGRAM
207a
PRE-EXECUTION PROCESSING PROGRAM
207b
ACTUAL EXECUTION PRO-CESSING PROGRAM
207c

FIG. 11

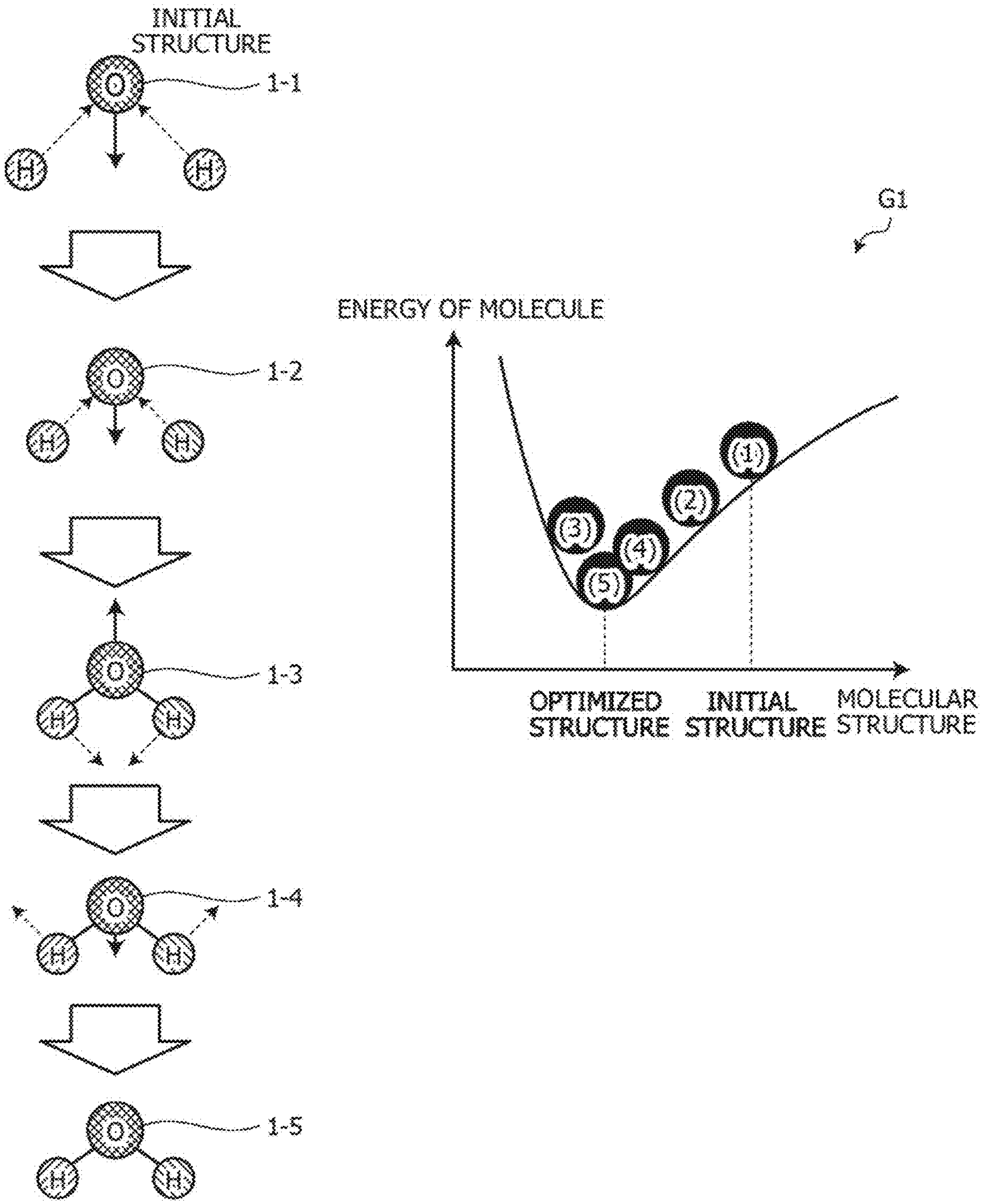

INITIAL STRUCTURE
O
H
H
1-1
1-2
1-3
1-4
1-5
G1
ENERGY OF MOLECULE
(1)
(2)
(3)
(4)
(5)
OPTIMIZED STRUCTURE
INITIAL STRUCTURE
MOLECULAR STRUCTURE

FIG. 12

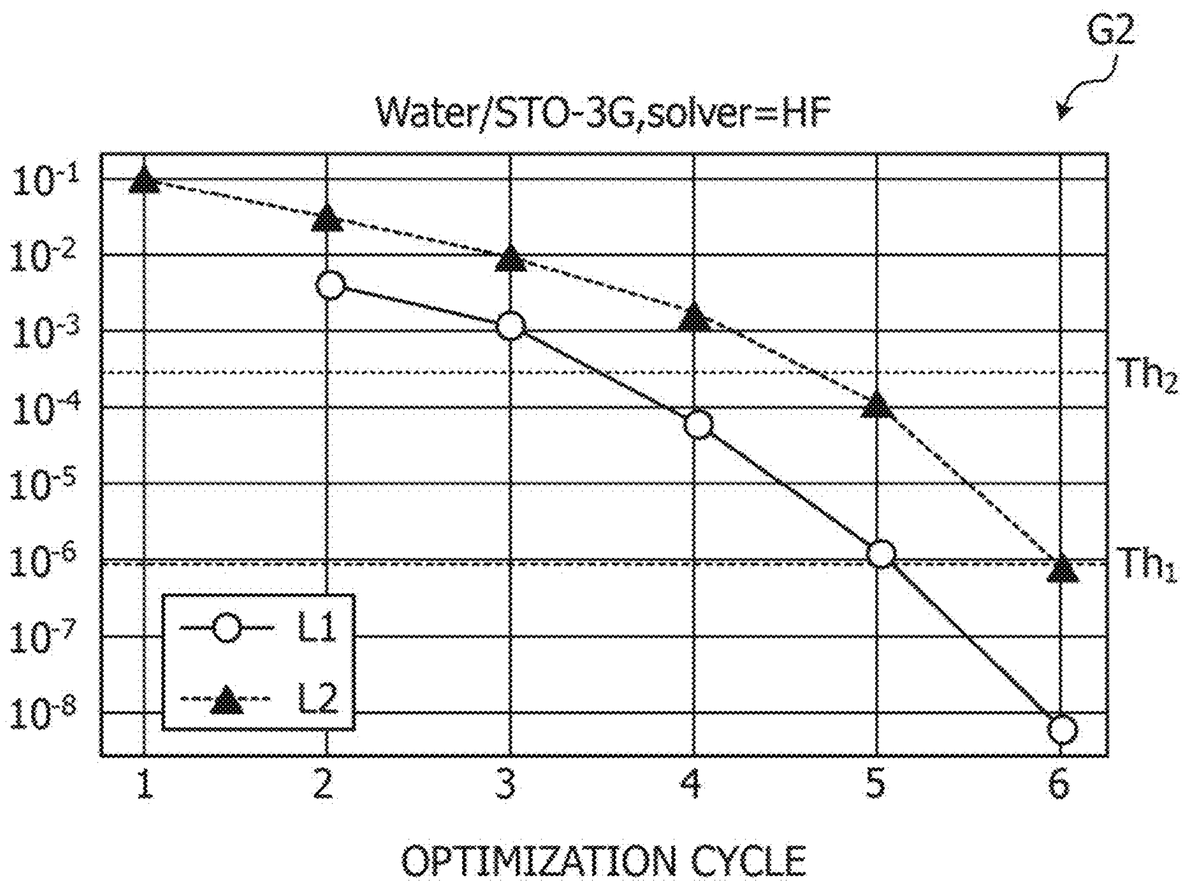

G2
Water/STO-3G,solver=HF
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
Th_2
Th_1
L1
L2
1
2
3
4
5
6
OPTIMIZATION CYCLE

FIG. 13

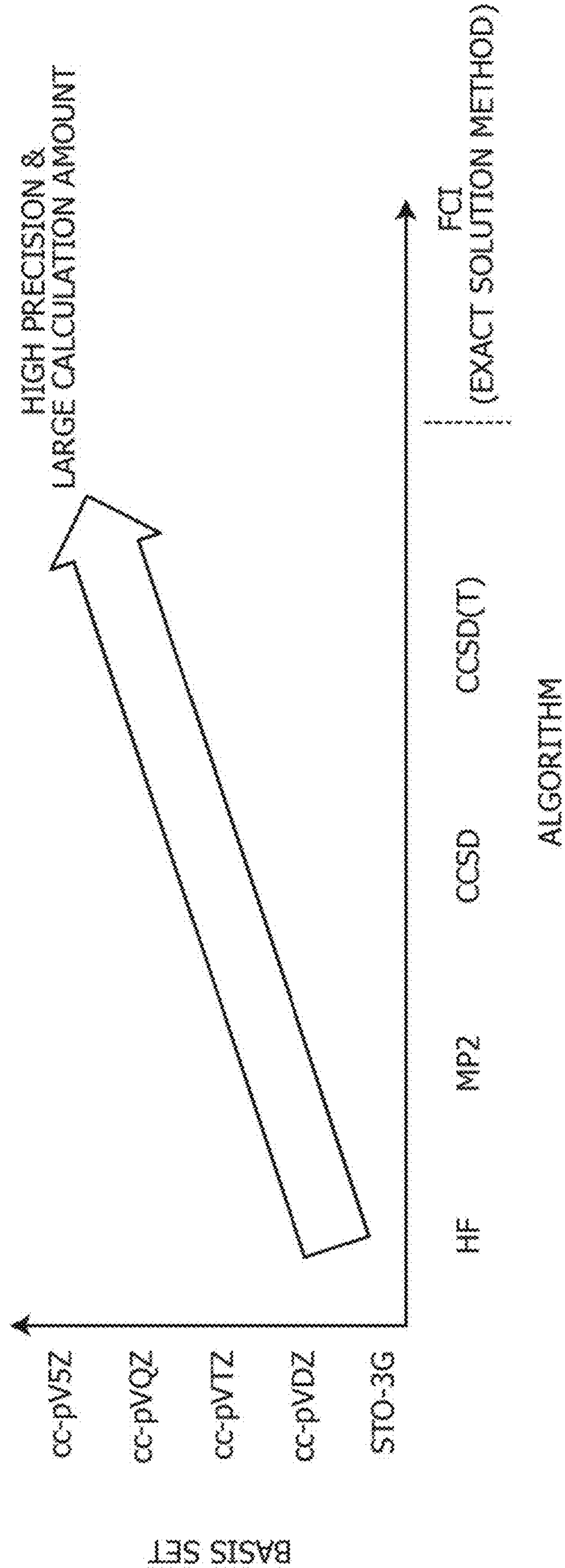
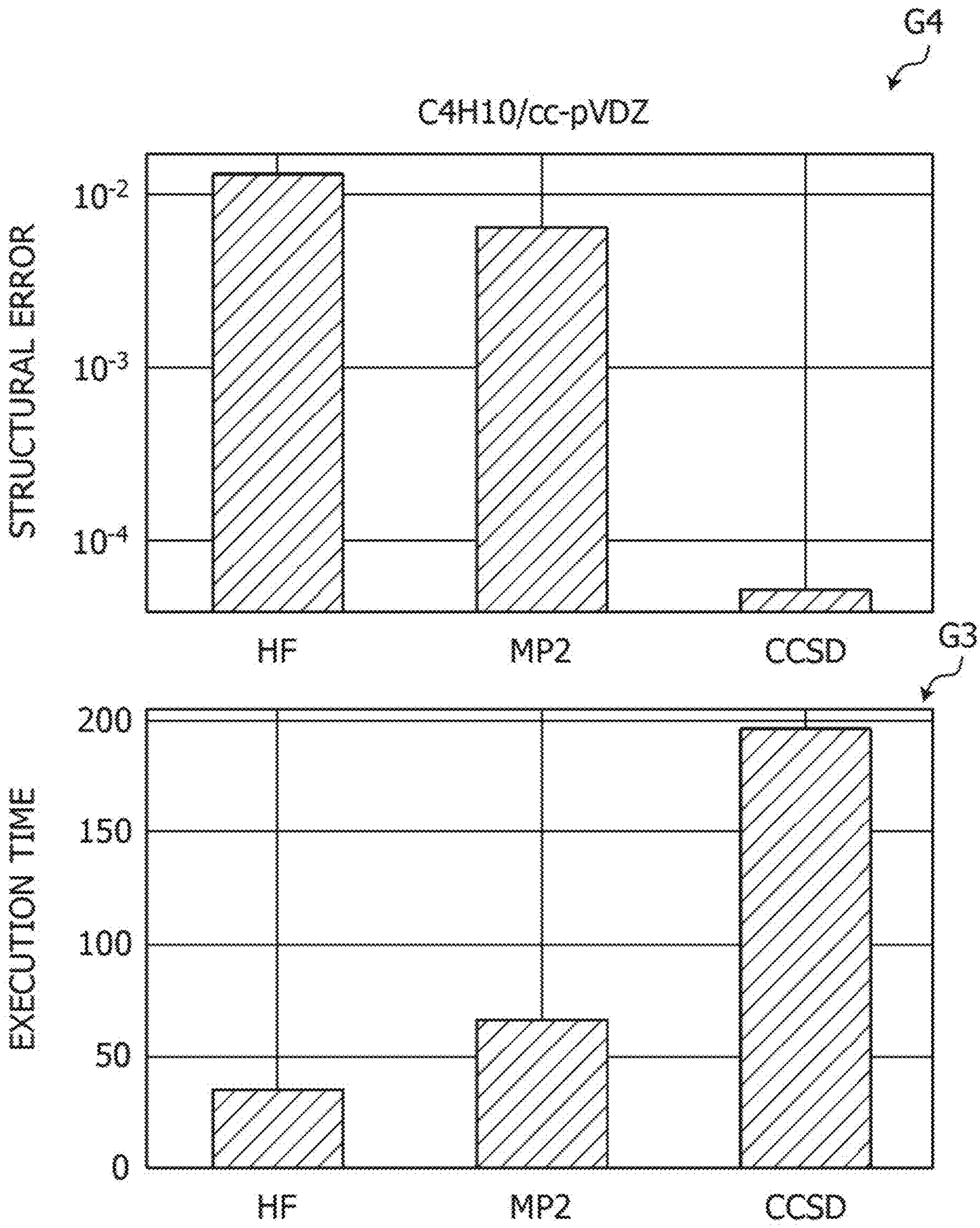

HIGH PRECISION &
LARGE CALCULATION AMOUNT
cc-pV5Z
cc-pVQZ
cc-pVTZ
cc-pVDZ
STO-3G
BASIS SET
HF
MP2
CCSD
CCSD(T)
FCI
(EXACT SOLUTION METHOD)
ALGORITHM

FIG. 14


G4
C4H10/cc-pVDZ
STRUCTURAL ERROR
10-2
10-3
10-4
HF
MP2
CCSD
G3
EXECUTION TIME
200
150
100
50
0
HF
MP2
CCSD

COMPUTER-READABLE RECORDING MEDIUM STORING ALGORITHM SELECTION PROGRAM, ALGORITHM SELECTION METHOD, AND INFORMATION PROCESSING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2024-29192, filed on Feb. 28, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The embodiment discussed herein is related to a computer-readable recording medium storing an algorithm selection program and the like.

BACKGROUND

[0003] As a method for analyzing the structure and properties of a molecule from an electronic state, there is quantum chemical calculation. Quantum chemical calculation is based on calculating the energy of a target molecule. By obtaining the energy of a molecule, various characteristics may be grasped, which may be used for drug discovery and development of new materials.

[0004] International Publication Pamphlet No. WO 2011/036952 is disclosed as related art.

SUMMARY

[0005] According to an aspect of the embodiments, a non-transitory computer-readable recording medium stores an algorithm selection program for causing a computer to execute a process including: in a case where a solution for structure optimization of a molecule is calculated using a quantum chemical calculation algorithm that repeatedly executes calculation for a plurality of cycles, executing a first quantum chemical calculation algorithm for a target molecule and calculating a nuclear gradient norm for each cycle; executing the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is equal to or more than a first threshold; and executing a second quantum chemical calculation algorithm of which execution time is longer than the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the first threshold.

[0006] The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.

[0007] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention.

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. **1** is a diagram for describing a technical problem related to algorithm switching;

[0009] FIG. **2** is a diagram illustrating the calculation precision of a nuclear gradient norm of lightweight algorithms;

[0010] FIG. **3**A is a diagram for describing processing of an information processing apparatus according to the present embodiment;

[0011] FIG. **3**B is a diagram illustrating the number of cycles taken for structure optimization by each algorithm;

[0012] FIG. **4** is a functional block diagram illustrating a configuration of the information processing apparatus according to the present embodiment;

[0013] FIG. **5** is a diagram illustrating an example of a data structure of algorithm candidate data;

[0014] FIG. **6** is a diagram illustrating an example of a data structure of an algorithm list;

[0015] FIG. **7** is a flowchart illustrating a processing procedure of the information processing apparatus according to the present embodiment;

[0016] FIG. **8** is a flowchart illustrating a processing procedure of algorithm list creation processing;

[0017] FIG. **9** is a flowchart illustrating a processing procedure of structure optimization processing;

[0018] FIG. **10** is a diagram illustrating an example of a hardware configuration of a computer that implements functions similar to those of the information processing apparatus of the present embodiment;

[0019] FIG. **11** is a diagram for describing molecular structure optimization;

[0020] FIG. **12** is a diagram illustrating the relationship between energy difference and nuclear gradient norm over the course of cycles;

[0021] FIG. **13** is a diagram for describing the relationship between precision level and calculation amount of quantum chemical calculation; and

[0022] FIG. **14** is a diagram for describing the relationship between execution time and precision related to structure optimization.

DESCRIPTION OF EMBODIMENTS

[0023] Representative processing performed in quantum chemical calculation includes structure optimization of molecules. In structure optimization, a stable structure of a molecule is obtained by molecular structure calculation. For example, in structure optimization, energy calculation and nuclear gradient calculation are repeatedly executed while changing the structure of a molecule, and when the energy difference or nuclear gradient norm falls below a threshold, convergence occurs.

[0024] FIG. **11** is a diagram for describing molecular structure optimization. In FIG. **11**, description is given using the water molecule "H2O" as an example. Graph G**1** indicates the energy of molecule in each structure of the molecule (water molecule). The horizontal axis of graph G**1** is an axis corresponding to molecular structure, and the vertical axis is an axis corresponding to energy of molecule. Hereinafter, energy of molecule will be referred to as "energy" as appropriate.

[0025] In structure optimization, the position of each atom is displaced little by little according to the vector of the force acting on each atom, and the structure is optimized little by little. For example, the initial structure of the molecule is molecular structure **1-1**. The position in graph G**1** corresponding to molecular structure **1-1** and the energy is (**1**). When molecular structure **1-1** is changed to molecular structure **1-2** by structure optimization, the position in graph G**1** corresponding to the molecular structure and energy is (**2**).

[0026] Subsequently, when molecular structure **1-2** is changed to molecular structure **1-3** by structure optimization, the position in graph G**1** corresponding to the molecular structure and energy is (**3**). When molecular structure **1-3** is changed to molecular structure **1-4**, the position in graph G**1** corresponding to the molecular structure and energy is (**4**). When molecular structure **1-4** is changed to molecular structure **1-5**, the position in graph G**1** corresponding to the molecular structure and energy is (**5**).

[0027] Molecular structure **1-5** has the minimum value of energy, and is the optimized structure.

[0028] FIG. **12** is a diagram illustrating the relationship between energy difference and nuclear gradient norm over the course of cycles. The horizontal axis of graph G**2** is an axis corresponding to cycle (optimization cycle), and the vertical axis corresponds to value (value of energy difference or value of nuclear gradient norm). Line L**1** indicates the relationship between cycle and energy difference. Line L**2** indicates the relationship between cycle and nuclear gradient norm.

[0029] For example, the processing from update of a molecular structure to calculation of energy and nuclear gradient of the updated molecular structure is one cycle. Hereinafter, the n-th cycle is referred to as cycle n. An energy difference indicates a difference between the energy of a molecular structure in the previous cycle and the energy of the molecular structure in the current cycle. A nuclear gradient norm is the length of a sum of vectors acting on each atom obtained from a calculation result of nuclear gradient. An energy difference may be calculated in two or more number of cycles. A nuclear gradient norm may be calculated in one or more number of cycles.

[0030] In graph G**2** illustrated in FIG. **12**, in cycle "6", the energy difference is less than a threshold Th_1, and the nuclear gradient norm is less than a threshold Th_2. For example, in cycle "6", the molecule has the optimized structure, and it is indicated that the number of cycles for reaching the optimized structure (the number of optimization cycles) is "6".

[0031] Next, the relationship between precision level and calculation amount of the quantum chemical calculation described above will be described. FIG. **13** is a diagram for describing the relationship between precision level and calculation amount of quantum chemical calculation. As illustrated in FIG. **13**, the precision level and calculation amount are determined according to the pair of basis set and algorithm.

[0032] A basis set is a set of functions used for describing the state of electrons in a molecule. For example, basis sets include STO-3G, cc-pVDZ, cc-pVTZ, cc-pVQZ, cc-pV5Z, and the like. The problem scale for handling molecules increases in the order of STO-3G, cc-pVDZ, cc-pVTZ, cc-pVQZ, and cc-pV5Z (STO-3G has the smallest problem scale).

[0033] An algorithm is a technique for analyzing the electronic structure of a molecule, and energy and nuclear gradient are obtained. An algorithm performs analysis using any of the basis functions included in a basis set. Algorithms include hartree fock method (HF), second-order moller-plesset method (MP2), coupled cluster method (CC method), and full configuration interaction method (FCI). CCSD is one of the CC methods, and SD is an abbreviation for "singles and doubles". CCSD(T) is one of the CC methods, and SD(T) is an abbreviation for "singles, doubles, and perturbative triples". The calculation amount increases and the precision basically increases in the order of HF, MP2, CCSD, CCSD(T), and FCI.

[0034] For example, when quantum chemical calculation is executed by a pair of a basis set with a large problem scale and an algorithm with a large calculation amount, the precision level is increased, but the execution time is increased. On the other hand, when quantum chemical calculation is executed by a pair of a basis set with a small problem scale and an algorithm with a small calculation amount, the precision level is lowered, but the execution time is shortened.

[0035] In the above-described related art, there is a trade-off relationship between execution time and precision related to structure optimization.

[0036] FIG. **14** is a diagram for describing the relationship between execution time and precision related to structure optimization. In FIG. **14**, as an example, the target molecule for structure optimization is "C4H10", and the algorithms are HF, MP2, and CCSD. The basis set used by each algorithm is "cc-pVDZ".

[0037] Graph G**3** illustrates the execution time for each algorithm. The vertical axis of graph G**3** is an axis corresponding to execution time, and the horizontal axis is an axis corresponding to algorithm. As illustrated in graph G**3**, the execution time increases in the order of HF, MP2, and CCSD.

[0038] Graph G**4** illustrates the structural error (precision) for each algorithm. The larger the structural error, the lower the precision. The vertical axis of graph G**4** is an axis corresponding to structural error, and the horizontal axis is an axis corresponding to algorithm. As illustrated in graph G**4**, the structural error decreases in the order of HF, MP2, and CCSD.

[0039] As illustrated in graphs G**3** and G**4**, it may be seen that an algorithm with a large calculation amount has to be used for high-precision structure optimization, and an algorithm with a larger calculation amount takes a longer time for structure optimization. The execution time is calculated by "execution time=(energy calculation time+nuclear gradient calculation time)×number of cycles", and the energy calculation time and nuclear gradient calculation time increase to the power of the molecular scale.

[0040] For example, it is desired to shorten the time taken for structure optimization.

[0041] In one aspect, an object of the present disclosure is to provide an algorithm selection program, an algorithm selection method, and an information processing apparatus capable of shortening the time taken for structure optimization.

[0042] Hereinafter, an embodiment of an algorithm selection program, an algorithm selection method, and an information processing apparatus disclosed in the present application will be described in detail with reference to the drawings. This disclosure is not limited by this embodiment.

Embodiment

[0043] As an approach for shortening the execution time of high-precision structure optimization, it is conceivable to use a lightweight algorithm in the first several cycles and switch to a high-precision algorithm in the optimal cycle. For example, there is an approach in which an initial structure of a molecule is brought close to an optimized structure first by using HF, MP2, or the like, and then the optimized structure is calculated by CCSD or the like.

[0044] A technical problem in a case of switching from a lightweight algorithm to a high-precision algorithm will be described. FIG. **1** is a diagram for describing a technical problem related to algorithm switching. For example, the lightweight algorithms are HF and MP2, and the high-precision algorithm is CCSD. The basis set used by each algorithm is "STO-3G", and the target molecule for structure optimization is "Caffeine".

[0045] First, graph G**5** will be described. Graph G**5** illustrates the execution time of structure optimization in a case where the lightweight algorithm is executed for x cycles (the number of cycles x) first, and then the algorithm is switched to the high-precision algorithm. This execution time is the total execution time of the execution time of the lightweight algorithm and the execution time of the high-precision algorithm. The horizontal axis of graph G**5** corresponds to number of cycles in which the lightweight algorithm is executed, and the vertical axis is an axis corresponding to execution time. T_1 is the execution time of structure optimization in a case where only the high-precision algorithm is used from the beginning.

[0046] Line L**3** in graph G**5** indicates the execution time in a case where CCSD is executed after HF is executed first for x cycles. Line L**4** in graph G**5** indicates the execution time in a case where CCSD is executed after MP2 is executed first for x cycles.

[0047] In graph G**5**, the number of cycles x for which the execution time of line L**3** is equal to or more than T_1 indicates that the execution time in the case where CCSD is executed after HF is executed first for x cycles is longer than the execution time in a case where CCSD is executed from the beginning. In graph G**5**, the number of cycles x for which the execution time of line L**4** is equal to or more than T_1 indicates that the execution time in the case where CCSD is executed after MP2 is executed first for x cycles is longer than the execution time in the case where CCSD is executed from the beginning.

[0048] For example, when attention is focused on the number of cycles "x=4" in graph G**5**, the execution time of line L**3** is the execution time equal to or more than T_1. This indicates that the execution time in the case where CCSD is executed after HF is executed first for four cycles is longer than the execution time in the case where CCSD is executed from the beginning. On the other hand, the execution time of line L**4** is the execution time of less than T_1. This indicates that the execution time in the case where CCSD is executed after MP2 is executed first for four cycles is shorter than the execution time in the case where CCSD is executed from the beginning.

[0049] Next, graph G**6** will be described. Graph G**6** illustrates the total number of cycles of structure optimization in the case where the lightweight algorithm is executed for x cycles (the number of cycles x) first, and then the algorithm is switched to the high-precision algorithm. This total number of cycles is the total number of cycles of the number of cycles of the lightweight algorithm and the number of cycles of the high-precision algorithm. The horizontal axis of graph G**6** corresponds to number of cycles in which the lightweight algorithm is executed, and the vertical axis is an axis corresponding to total number of cycles. C_1 is the total number of cycles in the case where only the high-precision algorithm is used.

[0050] Line L**3** in graph G**6** indicates the total number of cycles in the case where CCSD is executed after HF is executed first for x cycles. Line L**4** in graph G**6** indicates the execution time in the case where CCSD is executed after MP2 is executed first for x cycles.

[0051] In graph G**6**, the number of cycles x for which the total number of cycles of line L**3** is equal to or more than C_1 indicates that the total number of cycles in the case where CCSD is executed after HF is executed first for x cycles is larger than the total number of cycles in the case where CCSD is executed from the beginning. In graph G**6**, the number of cycles x for which the total number of cycles of line L**4** is equal to or more than C_1 indicates that the total number of cycles in the case where CCSD is executed after MP2 is executed first for x cycles is larger than the total number of cycles in the case where CCSD is executed from the beginning.

[0052] For example, when attention is focused on the number of cycles "x=4" in graph G**6**, the total number of cycles of line L**3** is the total number of cycles equal to or more than C_1. This indicates that the total number of cycles in the case where CCSD is executed after HF is executed first for four cycles is larger than the total number of cycles in the case where CCSD is executed from the beginning. On the other hand, the total number of cycles of line L**4** is substantially the same as C_1. This indicates that the total number of cycles in the case where CCSD is executed after MP2 is executed first for four cycles is substantially the same as the total number of cycles in the case where CCSD is executed from the beginning.

[0053] As described in graphs G**5** and G**6** of FIG. **1**, if the number of cycles in which the lightweight algorithm is used is too large, the total number of cycles increases and it is difficult to shorten the execution time. If the number of cycles in which the lightweight algorithm is used is too large, the precision may deteriorate. The point at which the total number of cycles begins to increase differs depending on the type of lightweight algorithm. Therefore, it is important to determine the number of cycles in which the lightweight algorithm is used so that the execution time taken for structure optimization is minimized.

[0054] The technical problem in the case of switching from a lightweight algorithm to a high-precision algorithm has been described above.

[0055] As described in graph G**2** of FIG. **12**, a nuclear gradient norm may be calculated from the first cycle. Attention is focused on the fact that since the nuclear gradient norm is about 10^{-2} to 10^{-1} in the first several cycles regardless of the algorithm, precise calculation may be performed even with a lightweight algorithm.

[0056] FIG. **2** is a diagram illustrating the calculation precision of a nuclear gradient norm of lightweight algorithms. Graph G**7** in FIG. **2** illustrates errors with respect to CCSD in a case where nuclear gradient norm calculation is performed using lightweight algorithms (HF and MP2) for each molecule. The horizontal axis of graph G**7** is an axis corresponding to types of molecules, and the vertical axis is an axis corresponding to error (error of nuclear gradient norm with respect to CCSD; hereinafter, simply referred to as error in the description of FIG. **2**).

[0057] In the example illustrated in FIG. **2**, the molecules used for structure optimization are LiH, O2, H2O, BeH2, NH3, CO2, HCl, CH4, C2H2, C2H4, C2H6, C3H4, C3H6, C3H8, C4H6, C4H8, and C4H10.

[0058] For example, in a case where HF is used as the algorithm, the maximum error among the errors for the respective molecules is "0.2". In a case where MP2 is used as the algorithm, the maximum error among the errors for the respective molecules is "0.03". As long as the nuclear gradient norm is larger than these maximum errors, calculation may be executed precisely even when a lightweight algorithm is used.

[0059] For example, in a case where structure optimization is performed, calculation may be executed precisely even when HF is used in the cycle during which the nuclear gradient norm is "0.2" or more. Similarly, in the case where structure optimization is performed, calculation may be executed precisely even when MP2 is used in the cycle during which the nuclear gradient norm is "0.03" or more.

[0060] The information processing apparatus according to the present embodiment focuses on the maximum error of nuclear gradient norm described in FIG. **2**, and switches the algorithm in the process of executing structure optimization, thereby making it possible to shorten the execution time of high-precision structure optimization. In the following description, the information processing apparatus according to the present embodiment is referred to as an "information processing apparatus **100**".

[0061] FIG. **3**A is a diagram for describing processing of the information processing apparatus according to the present embodiment. In the present embodiment, a case where the algorithms HF, MP2, and CCSD are switched stepwise will be described. The basis set used by each algorithm is "STO-3G", and the target molecule for structure optimization is "Caffeine". The information processing apparatus **100** performs actual execution processing after pre-execution processing is performed.

[0062] The pre-execution processing executed by the information processing apparatus **100** will be described. The information processing apparatus **100** selects the algorithm "HF" and the basis set "STO-3G", and executes structure optimization for the target molecule. This specifies the nuclear gradient norm for each cycle, which is calculated with the algorithm "HF" and the basis set "STO-3G". For example, the nuclear gradient norm for each cycle is illustrated by line L**5** in graph G**8**. The pair of algorithm and basis set is denoted as "HF/STO-3G".

[0063] The vertical axis of graph G**8** is an axis corresponding to nuclear gradient norm obtained with HF/STO-3G, and the horizontal axis is an axis corresponding to the number of cycles in which HF/STO-3G is executed. The example illustrated in FIG. **3**A indicates that the optimized structure is specified with HF/STO-3G during pre-execution, with the number of cycles "10".

[0064] Threshold Th_{HF} illustrated in graph G**8** is the maximum error of HF described in FIG. **2**. Threshold Th_{MP2} illustrated in graph G**8** is the maximum error of MP2 described in FIG. **2**.

[0065] The pre-execution processing executed by the information processing apparatus **100** has been described above.

[0066] The actual execution processing executed by the information processing apparatus **100** will be described. The information processing apparatus **100** selects an algorithm to be used for each cycle based on the result of the pre-execution processing (line L**5** in graph G**8**) and threshold Th_{HF} and threshold Th_{MP2}.

[0067] The information processing apparatus **100** selects the algorithm "HF" in the cycle period in which the nuclear gradient norm obtained with HF/STO-3G is equal to or more than threshold ThHF. The information processing apparatus **100** selects the algorithm "MP2" in the cycle period in which the nuclear gradient norm obtained with HF/STO-3G is less than threshold Th_{HF} and equal to or more than threshold Th_{MP2}. The information processing apparatus **100** selects the algorithm "CCSD" in the cycle period in which the nuclear gradient norm obtained with HF/STO-3G is less than threshold Th_{MP2}. The basis set to be used is designated by a user (it may be other than STO-3G).

[0068] In the example illustrated in FIG. **3**A, the information processing apparatus **100** selects "HF" in the first cycle, selects "MP2" in the second to fifth cycles, and selects "CCSD" in the sixth and subsequent cycles.

[0069] The information processing apparatus **100** executes structure optimization for the molecule based on the selected result. For example, the information processing apparatus **100** executes energy calculation and nuclear gradient calculation by "HF" in the first cycle. The information processing apparatus **100** executes energy calculation and nuclear gradient calculation by "MP2" in the second to fifth cycles. The information processing apparatus **100** executes energy calculation and nuclear gradient calculation by "CCSD" in the sixth and subsequent cycles.

[0070] As described above, the information processing apparatus **100** according to the present embodiment selects an algorithm to be used for each cycle based on the result of the pre-execution processing (line L**5** in graph G**8**) and threshold Th_{HF} and threshold Th_{MP2}, and executes structure optimization based on the selection result. As a result, a lightweight algorithm may be selected in a cycle period in which precise calculation may be performed even with a lightweight algorithm, and thereafter, a high-precision algorithm may be selected. For example, the time taken for structure optimization may be shortened while maintaining precision.

[0071] FIG. **3**B is a diagram illustrating the number of cycles taken for structure optimization by each algorithm. In FIG. **3**B, as an example, the target algorithms are HF, MP2, CCSD, and CCSD(T). The horizontal axis of graph G**9** in FIG. **3**B is an axis corresponding to type of molecule, and the vertical axis is an axis corresponding to the number of cycles for structure optimization. The number of cycles for structure optimization is the number of cycles taken for completion of structure optimization.

[0072] As illustrated in FIG. **3**B, it may be seen that the number of cycles for structure optimization of other algorithms (MP2, CCSD, and CCSD(T)) is not significantly different from the number of cycles for structure optimization of HF. From this, as described in FIG. **3**A, it may be said that even if HF is replaced with another algorithm, the number of cycles for structure optimization does not change greatly.

[0073] Next, a configuration example of the information processing apparatus **100** of the present embodiment will be described. FIG. **4** is a functional block diagram illustrating a configuration of the information processing apparatus according to the present embodiment. As illustrated in FIG. **4**, the information processing apparatus **100** includes a communication unit **110**, an input unit **120**, a display unit **130**, a storage unit **140**, and a control unit **150**.

[0074] The communication unit **110** executes data communication with an external device or the like via a network. The communication unit **110** is a network interface card (NIC) or the like. For example, the communication unit **110** may acquire, from the external device or the like, molecular initial structure data **141**, basis set data **142**, algorithm candidate data **144**, and the like.

[0075] The input unit **120** is an input device that inputs various types of information to the control unit **150** of the information processing apparatus **100**. For example, the input unit **120** corresponds to a keyboard, a mouse, a touch panel, or the like.

[0076] The display unit **130** is a display device that displays information output from the control unit **150**.

[0077] The storage unit **140** includes the molecular initial structure data **141**, the basis set data **142**, a nuclear gradient norm list **143**, the algorithm candidate data **144**, an algorithm list **145**, and optimized structure data **146**. The storage unit **140** is a memory or the like.

[0078] The molecular initial structure data **141** is data of an initial structure of a target molecule of structure optimization.

[0079] The basis set data **142** includes basis sets such as STO-3G, cc-pVDZ, cc-pVTZ, cc-pVQZ, and cc-pV5Z. The basis set used in structure optimization is designated in advance by a user.

[0080] The nuclear gradient norm list **143** is a list generated by the pre-execution processing, and the nuclear gradient norm calculated in each cycle is set therein. For example, the nuclear gradient norm list **143** is information corresponding to line L**5** in graph G**8** described in FIG. **3**A.

[0081] In the algorithm candidate data **144**, algorithms to be used in a case where structure optimization is performed for a molecule are set. For example, in the algorithm candidate data **144**, types of algorithms and thresholds are associated with each other.

[0082] FIG. **5** is a diagram illustrating an example of a data structure of algorithm candidate data. As illustrated in FIG. **5**, in the algorithm candidate data **144**, algorithms and thresholds are associated with each other. As the algorithms, the types of algorithms described above are set. A threshold is the maximum error corresponding to each algorithm. Description of the maximum error is similar to the description given in FIG. **3**A. For example, the order of the elements (records) of the algorithm candidate data is descending order of thresholds from the top.

[0083] In the algorithm list **145**, an algorithm used for each cycle is set. FIG. **6** is a diagram illustrating an example of a data structure of the algorithm list. In the example illustrated in FIG. **6**, a cycle number and the algorithm used in the corresponding cycle are associated with each other.

[0084] The optimized structure data **146** is an execution result of structure optimization for the molecular initial structure data **141**.

[0085] Next, description proceeds to that of the control unit **150**. The control unit **150** includes an acquisition unit **151**, a pre-execution processing unit **152**, and an actual execution processing unit **153**. The control unit **150** is a central processing unit (CPU), a graphics processing unit (GPU), or the like.

[0086] The acquisition unit **151** acquires the molecular initial structure data **141**, the basis set data **142**, the algorithm candidate data **144**, and the like via the communication unit **110**. The acquisition unit **151** may acquire the molecular initial structure data **141**, the basis set data **142**, the algorithm candidate data **144**, and the like from the input unit **120**.

[0087] The acquisition unit **151** stores the acquired molecular initial structure data **141**, basis set data **142**, algorithm candidate data **144**, and the like in the storage unit **140**.

[0088] The pre-execution processing unit **152** performs the pre-execution processing described in FIG. **3**A. For example, the pre-execution processing unit **152** executes structure optimization for the molecular initial structure data **141** with "HF/STO-3G".

[0089] The pre-execution processing unit **152** calculates energy and nuclear gradient and calculates a nuclear gradient norm for each cycle in the process of structure optimization. The pre-execution processing unit **152** registers the relationship between the number of cycles and the nuclear gradient norm in the nuclear gradient norm list **143**.

[0090] Other description related to the pre-execution processing unit **152** is similar to the description of the pre-execution processing described in FIG. **3**A.

[0091] The actual execution processing unit **153** performs the actual execution processing described in FIG. **3**A. For example, the actual execution processing unit **153** creates the algorithm list **145** by selecting an algorithm to be used for each cycle based on the nuclear gradient norm list **143** and the algorithm candidate data **144**. The basis set used in the actual execution processing is designated in advance by a user.

[0092] The actual execution processing unit **153** compares the nuclear gradient norm of each cycle in the nuclear gradient norm list **143** with the threshold in the algorithm candidate data **144**. The actual execution processing unit **153** selects the algorithm "HF" in a cycle period in which the nuclear gradient norm is equal to or more than threshold Th_{HF} (for example, the first cycle).

[0093] The actual execution processing unit **153** selects the algorithm "MP2" in a cycle period in which the nuclear gradient norm is less than threshold Th_{HF} and equal to or more than threshold Th_{MP2} (for example, the second cycle to the fifth cycle). The actual execution processing unit **153** selects the algorithm "CCSD" in a cycle period in which the nuclear gradient norm is less than threshold Th_{MP2} (for example, the sixth cycle to the eleventh cycle).

[0094] By the actual execution processing unit **153** executing the above processing, the algorithm list **145** illustrated in FIG. **6** is created.

[0095] The actual execution processing unit **153** selects an algorithm for each number of cycles based on the created algorithm list **145**, and executes structure optimization for the molecular initial structure data **141**. For example, in structure optimization, the actual execution processing unit **153** selects and executes the algorithm "HF" in the first cycle. The actual execution processing unit **153** selects and executes the algorithm "MP**2**" in the second cycle to the fifth cycle. The actual execution processing unit **153** selects and executes the algorithm "CCSD" in the sixth cycle to the eleventh cycle.

[0096] The actual execution processing unit **153** stores the execution result of structure optimization in the storage unit **140** as the optimized structure data **146**. The actual execution processing unit **153** may output the optimized structure data **146** to the display unit **130** for display.

[0097] Next, an example of a processing procedure of the information processing apparatus **100** according to the present embodiment will be described. FIG. **7** is a flowchart illustrating a processing procedure of the information processing apparatus according to the present embodiment. The acquisition unit **151** of the information processing apparatus **100** acquires the molecular initial structure data **141** and the algorithm candidate data **144** (step S**101**).

[0098] The pre-execution processing unit **152** of the information processing apparatus **100** pre-executes structure optimization for a target molecule with HF/STO-3G, and creates the nuclear gradient norm list **143** (step S**102**).

[0099] The actual execution processing unit **153** of the information processing apparatus **100** executes algorithm list creation processing (step S**103**). The actual execution processing unit **153** executes structure optimization processing (step S**104**).

[0100] The actual execution processing unit **153** outputs the optimized structure data **146** (step S**105**).

[0101] Next, an example of a processing procedure of the algorithm list creation processing described in step S**103** of FIG. **7** will be described. FIG. **8** is a flowchart illustrating a processing procedure of the algorithm list creation processing. As illustrated in FIG. **8**, the actual execution processing unit **153** of the information processing apparatus **100** initializes the algorithm list **145** to be empty (step S**201**).

[0102] The actual execution processing unit **153** takes out the first element of the algorithm candidate data **144** (step S**202**). The actual execution processing unit **153** takes out the first element of the nuclear gradient norm list **143** (step S**203**).

[0103] When the nuclear gradient norm is larger than the threshold of the corresponding algorithm (step S**204**, Yes), the actual execution processing unit **153** proceeds to step S**207**. On the other hand, when the nuclear gradient norm is not larger than the threshold of the corresponding algorithm (step S**204**, No), the actual execution processing unit **153** proceeds to step S**205**.

[0104] When the algorithm candidate data **144** is empty (step S**205**, Yes), the actual execution processing unit **153** proceeds to step S**207**. On the other hand, when the algorithm candidate data **144** is not empty (step S**205**, No), the actual execution processing unit **153** takes out the first element of the algorithm candidate data **144** (step S**206**), and proceeds to step S**207**.

[0105] The actual execution processing unit **153** adds the corresponding algorithm to the end of the algorithm list **145** (step S**207**). When the nuclear gradient norm list is not empty (step S**208**, No), the actual execution processing unit **153** proceeds to step S**203**. On the other hand, when the nuclear gradient norm list is empty (step S**208**, Yes), the actual execution processing unit **153** ends the algorithm list creation processing.

[0106] Next, an example of a processing procedure of the structure optimization processing described in step S**104** of FIG. **7** will be described. FIG. **9** is a flowchart illustrating a processing procedure of the structure optimization processing. As illustrated in FIG. **9**, the actual execution processing unit **153** of the information processing apparatus **100** takes out the first element of the algorithm list **145** (step S**301**).

[0107] The actual execution processing unit **153** calculates energy and nuclear gradient with the corresponding algorithm (step S**302**). The actual execution processing unit **153** changes the molecular structure (step S**303**).

[0108] When a convergence condition is not satisfied (step S**304**, No), the actual execution processing unit **153** proceeds to step S**305**. When the algorithm list **145** is empty (step S**305**, Yes), the actual execution processing unit **153** proceeds to step S**302**. On the other hand, when the algorithm list **145** is not empty (step S**305**, No), the actual execution processing unit **153** proceeds to step S**301**.

[0109] When the convergence condition is satisfied (step S**304**, Yes), the actual execution processing unit **153** outputs the molecular structure at the current stage as the optimized structure data **146** (step S**306**).

[0110] Next, the effects of the information processing apparatus **100** according to the present embodiment will be described. The information processing apparatus **100** selects an algorithm to be used for each cycle based on a result of pre-execution processing and a threshold set for each algorithm, and executes structure optimization based on the selection result. As a result, a lightweight algorithm may be selected in a cycle period in which precise calculation may be performed even with a lightweight algorithm, and thereafter, a high-precision algorithm may be selected. For example, the time taken for structure optimization may be shortened while maintaining precision.

[0111] The information processing apparatus **100** creates an algorithm list in which an algorithm to be used for each cycle is selected based on a result of pre-execution processing and a threshold set for each algorithm. The information processing apparatus **100** may execute structure optimization by switching from a lightweight algorithm to a high-precision algorithm in a stepwise manner in accordance with the algorithm list.

[0112] In the present embodiment, description has been given with the basis set used in each algorithm in the pre-execution processing as "STO-3G", but the present disclosure is not limited thereto, and a different basis set may be used for each algorithm. However, it is desirable that a basis set of a small scale is selected in order to shorten the execution time of pre-execution processing.

[0113] Next, an example of a hardware configuration of a computer that implements functions similar to those of the information processing apparatus **100** described above will be described. FIG. **10** is a diagram illustrating an example of a hardware configuration of a computer that implements functions similar to those of the information processing apparatus of the present embodiment.

[0114] As illustrated in FIG. **10**, a computer **200** includes a CPU **201** that executes various types of arithmetic processing, an input device **202** that receives input of data from a user, and a display **203**. The computer **200** includes a communication device **204** that exchanges data with an external device or the like via a wired or wireless network, and an interface device **205**. The computer **200** includes a random-access memory (RAM) **206** that temporarily stores various types of information, and a hard disk device **207**. The devices **201** to **207** are coupled to a bus **208**.

[0115] The hard disk device **207** includes an acquisition program **207*a***, a pre-execution processing program **207*b***, and an actual execution processing program **207*c***. The CPU **201** reads the programs **207*a*** to **207*c*** and loads them to the RAM **206**.

[0116] The acquisition program **207*a*** functions as an acquisition process **206*a***. The pre-execution processing program **207*b*** functions as a pre-execution processing process

206*b*. The actual execution processing program **207***c* functions as an actual execution processing process **206***c*.

[0117] The processing of the acquisition process **206***a* corresponds to the processing of the acquisition unit **151**. The processing of the pre-execution processing process **206***b* corresponds to the processing of the pre-execution processing unit **152**. The processing of the actual execution processing process **206***c* corresponds to the processing of the actual execution processing unit **153**.

[0118] The programs **207***a* to **207***c* do not have be stored in the hard disk device **207** from the beginning. For example, each program is stored in a "portable physical medium", such as a flexible disk (FD), a compact disc read-only memory (CD-ROM), a Digital Versatile Disc (DVD), a magneto-optical disk, or an integrated circuit (IC) card, to be inserted into the computer **200**. The computer **200** may read and execute the programs **207***a* to **207***c*.

[0119] All examples and conditional language provided herein are intended for the pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventor to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although one or more embodiments of the present invention have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

What is claimed is:

1. A non-transitory computer-readable recording medium storing an algorithm selection program for causing a computer to execute a process comprising:

in a case where a solution for structure optimization of a molecule is calculated using a quantum chemical calculation algorithm that repeatedly executes calculation for a plurality of cycles, executing a first quantum chemical calculation algorithm for a target molecule and calculating a nuclear gradient norm for each cycle;

executing the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is equal to or more than a first threshold; and

executing a second quantum chemical calculation algorithm of which execution time is longer than the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the first threshold.

2. The non-transitory computer-readable recording medium according to claim **1**, wherein the computer is further caused to execute a process of executing the second quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the first threshold and equal to or more than a second threshold, and executing a third quantum chemical calculation algorithm of which execution time is longer than the second quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the second threshold.

3. The non-transitory computer-readable recording medium according to claim **1**, wherein the second quantum chemical calculation algorithm has higher precision than the first quantum chemical calculation algorithm.

4. The non-transitory computer-readable recording medium according to claim **2**, wherein the third quantum chemical calculation algorithm has higher precision than the second quantum chemical calculation algorithm.

5. The non-transitory computer-readable recording medium according to claim **2**, wherein the computer is further caused to execute a process of selecting a quantum chemical calculation algorithm to be executed for the each cycle from the first quantum chemical calculation algorithm, the second quantum chemical calculation algorithm, and the third quantum chemical calculation algorithm based on the nuclear gradient norm for each cycle, the first threshold, the second threshold, and the third threshold.

6. An algorithm selection method for causing a computer to execute a process comprising:

in a case where a solution for structure optimization of a molecule is calculated using a quantum chemical calculation algorithm that repeatedly executes calculation for a plurality of cycles, executing a first quantum chemical calculation algorithm for a target molecule and calculating a nuclear gradient norm for each cycle;

executing the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is equal to or more than a first threshold; and

executing a second quantum chemical calculation algorithm of which execution time is longer than the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the first threshold.

7. The algorithm selection method according to claim **6**, wherein the computer is further caused to execute a process of executing the second quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the first threshold and equal to or more than a second threshold, and executing a third quantum chemical calculation algorithm of which execution time is longer than the second quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the second threshold.

8. The algorithm selection method according to claim **6**, wherein the second quantum chemical calculation algorithm has higher precision than the first quantum chemical calculation algorithm.

9. The algorithm selection method according to claim **7**, wherein the third quantum chemical calculation algorithm has higher precision than the second quantum chemical calculation algorithm.

10. The algorithm selection method according to claim **7**, wherein the computer is further caused to execute a process of selecting a quantum chemical calculation algorithm to be executed for the each cycle from the first quantum chemical calculation algorithm, the second quantum chemical calculation algorithm, and the third quantum chemical calculation algorithm based on the nuclear gradient norm for each cycle, the first threshold, the second threshold, and the third threshold.

11. An information processing apparatus comprising:

a memory; and

a processor coupled to the memory and configured to:

in a case where a solution for structure optimization of a molecule is calculated using a quantum chemical calculation algorithm that repeatedly executes calculation for a plurality of cycles, execute a first quantum chemical calculation algorithm for a target molecule and calculate a nuclear gradient norm for each cycle;

execute the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is equal to or more than a first threshold; and

execute a second quantum chemical calculation algorithm of which execution time is longer than the first quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the first threshold.

12. The information processing apparatus according to claim **11**, wherein to the processor executes the second quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the first threshold and equal to or more than a second threshold, and executes a third quantum chemical calculation algorithm of which execution time is longer than the second quantum chemical calculation algorithm for a cycle in which the nuclear gradient norm is less than the second threshold.

13. The information processing apparatus according to claim **11**, wherein the second quantum chemical calculation algorithm has higher precision than the first quantum chemical calculation algorithm.

14. The information processing apparatus according to claim **12**, wherein the third quantum chemical calculation algorithm has higher precision than the second quantum chemical calculation algorithm.

15. The information processing apparatus according to claim **12**, wherein the processor selects a quantum chemical calculation algorithm to be executed for the each cycle from the first quantum chemical calculation algorithm, the second quantum chemical calculation algorithm, and the third quantum chemical calculation algorithm based on the nuclear gradient norm for each cycle, the first threshold, the second threshold, and the third threshold.

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